

ICIC 2026 Paper # 2926 Decision Notification

Dear Author(s),

Congratulations! On behalf of the ICIC 2026 International Program Committee, we are very pleased to inform you that your paper:

Paper ID: 2926

Author(s): shengjie fan, le zhang, ming xiao

Title: Propensity-Weighted Modified Logistic Regression for Robust PU Learning under Structural Label Noise

Presentation: **Oral**

has been accepted for presentation at the 2026 International Conference on Intelligent Computing, which has been selected into the following Springer –Nature series:

● Lecture Notes in Artificial Intelligence (LNAI)

This year we received a large volume of paper submissions from **21** countries and regions, and each paper received averagely **3.6** review reports, only a limited number of papers (less than the acceptance rate of 40%) are accepted while a large number of high-quality papers have to be rejected due to the space limitation.

IMPORTANT NOTES:

- Please note that the papers submitted to the ICIC conference should not have been submitted to other conferences at the same time. If multiple submissions were found, the paper will be directly excluded from the conference proceedings. In the meantime, such multiple submission information will be passed to the other involved conferences.
- From previous ICIC2006 & ICIC2009 & ICIC2011 & ICIC2025 proceedings, it can be found that there are several **severe cases of plagiarism**. So we are kindly reminding all the authors to stipulate your academic behavior, and cherish and protect your own reputation as well as our ICIC reputation. We hope not to see this phenomenon in future ICIC academic event. We shall try our best to take every measure to stop this misconduct.

To include your paper in the proceedings to be published by Springer-Nature, here are the steps you must follow:

1. Please see the attached REVIEWERS' COMMENTS for your paper, which are intended to help you to improve your paper for final publication. The listed comments should be fully addressed, as acceptance is conditional on appropriate response to the requirements and comments, which only need to be stated in the final Paper. Unsatisfactory final paper submission may be directly rejected by the editor.

2. Please take a look at Proceedings Templates (论文排版格式) in the link (<http://www.ic-icc.cn/2026/Registration.php>) to obtain the information for preparing the final version of your paper. Note that for your camera-ready paper, you should first write your FULL given Names, and then write your FULL Family names, which is an international convention. In addition, the paper length of **12** pages for EACH paper in Springer format will be strictly enforced. If your paper is over the stipulated Maximum length, you will have to pay the

over-length page fee at \$200 USD or ¥1540 RMB (\$200*7.7) per extra page.

3. For each paper/poster, **Plagiarism Detection Rate must be less than 15%**. Otherwise, your paper will be excluded into the proceedings. So each camera-ready paper/poster has to be detected by any Plagiarism detection software before you are uploading to our online system.

When you have completed revising your paper and are ready to submit it, please go to the link http://www.ic-icc.cn/icg/fir_editor.php?identity=Rgistration to submit your final camera-ready paper. On this page you will need to use the following account and password:

Account: 2023141470277@stu.scu.edu.cn

Password: Fan12345@@

Final papers MUST be submitted before **May 12, 2026**. The paper must be re-submitted even if the reviewers indicated that no changes are required.

3. For each paper/poster, at least one of the authors for EACH paper must register for the conference. The deadline for author registration is **May 12, 2026**.

Particularly, please write **ICIC2026 + Paper ID Number** on the **BANK** transfer receipt when you are paying for a registration fee for your paper.

4. To be published in the conference proceedings, all accepted papers must be accompanied by a signed copyright form by downloading it from the **"Camera-ready Submission"** section of your personal account. Before submitting your final camera-ready paper, on the copyright form please write your Paper Title, Authors List, etc. and upload its scanned copy (in PDF format) together with the electronic bank transfer form (in PDF format) by <http://www.ic-icc.cn/2026/Registration.php>.

5. Hotel reservation information is available on the ICIC 2026 website very soon. Please keep an eye on <http://www.ic-icc.cn/2026/What's%20New.php> and book your room as early as possible!

If you require a visa to enter China please send a request for an invitation letter to Prashan Premaratne, University of Wollongong, Australia at prashan@uow.edu.au. Thank you for participating in ICIC2026. We look forward to seeing you in Toronto, Canada, July 22-26, 2026.

Thank you very much for your support to ICIC 2026.

Sincerely,

ICIC2026 Program Committee

REVIEWERS' COMMENTS

REVIEW NO. 1 (Turn 1)

Originality: 3.5

Significance of topic: 4

Technical quality: 3

Relevance to ICIC 2026: 4

Presentation: 4

Overall rating: 3.7

Reviewer's expertise on the topic:

Comments to the authors:

This paper investigates the problem of Positive-Unlabeled PU learning under structural label noise. The authors critically examine the failure modes of foundational linear models and introduce a Propensity-weighted Modified Logistic Regression P-MLR model. By parameterizing the propensity score, P-MLR achieves an end-to-end decoupling of the true class distribution and feature-dependent labeling bias. Weaknesses and Questions: The experimental evaluation only compares P-MLR against traditional models. Although the authors mention recent SOTA SAR-PU algorithms e.g., PUE based on causal inference in the introduction, none are included in the baseline comparison. A quantitative comparison with at least one recent advanced model is necessary to fully justify the performance-efficiency trade-off. If, in real-world medical scenarios, the structural noise does not follow a simple monotonic mapping mechanism e.g., larger feature norms do not necessarily lead to a higher likelihood of misdiagnosis, does the currently employed single-layer parameterized network possess sufficient representational capacity?

REVIEW NO. 2 (Turn 1)

Originality: 4

Significance of topic: 4

Technical quality: 3

Relevance to ICIC 2026: 4

Presentation: 3.5

Overall rating: 3.7

Reviewer's expertise on the topic:

Comments to the authors:

This paper addresses the PU learning problem under structural label noise, and innovatively integrates the SAR assumption into a lightweight linear model to propose the P-MLR method. It conforms to practical application scenarios and the experimental verification is relatively sufficient. After reading I still have the following questions and modification suggestions: 1.The proposed P-MLR method claims to achieve unbiased estimation of the positive class, is there any theoretical proof or justification for this? 2.The experimental section lacks key details for reproducibility, such as the training-validation set partitioning strategy and the number of model training epochs. There are some standardization issues in the paper. For example, SAR first appears in the abstract without its full English name.

REVIEW NO. 3 (Turn 1)

Originality: 3.5

Significance of topic: 4

Technical quality: 3.5

Relevance to ICIC 2026: 3.5

Presentation: 3

Overall rating: 3.5

Reviewer's expertise on the topic:

Comments to the authors:

Strength: The paper addresses a highly practical and important problem in PU learning: the robustness of lightweight linear models under real-world feature-dependent structural noise SAR assumption, which is a critical pain point for deploying PU models in high-risk

domains like medical diagnosis. Weaknesses Revision Suggestions: The method is only verified on tabular medical datasets. It is suggested to supplement experiments on other data types to verify the generalization ability of P-MLR in different PU learning scenarios, expanding the application scope of the method.

REVIEW NO. 4 (Turn 2)

Originality: 3.5

Significance of topic: 4

Technical quality: 4

Relevance to ICIC 2026: 4

Presentation: 4

Overall rating: 3.9

Reviewer's expertise on the topic:

Comments to the authors:

I have carefully read the authors' rebuttal and the revised manuscript. 1、The inclusion of the recent SOTA model PUE significantly elevates the quality of the empirical evaluation. The nuanced discussion added in Sections 4.1 and 4.2 successfully justifies the performance-efficiency trade-off. Demonstrating that P-MLR can match or even outperform complex deep learning models like PUE on structured tabular data. 2、Openly acknowledging the limitations of a single-layer linear mapping in capturing non-monotonic real-world clinical biases strengthens the paper's integrity. Adding this to the Limitations and Future Work section Section 4.3, along with the proposed future direction of using an MLP GAM for the propensity module, is a satisfactory resolution.

REVIEW NO. 5 (Turn 2)

Originality: 3.5

Significance of topic: 4

Technical quality: 3.5

Relevance to ICIC 2026: 3.5

Presentation: 3.5

Overall rating: 3.6

Reviewer's expertise on the topic:

Comments to the authors:

Thank you for the detailed response and sufficient supplementary experiments. As suggested, the authors have extended the validation datasets to cover image DIGITS, text SPAM_SMS, and non-medical structured tabular Spambase data modalities, which comprehensively verifies the generalization ability of P-MLR across diverse PU learning scenarios and expands the practical application scope of the proposed method.

REVIEW NO. 6 (Turn 2)

Originality: 4

Significance of topic: 4

Technical quality: 4

Relevance to ICIC 2026: 4

Presentation: 4

Overall rating: 4

Reviewer's expertise on the topic:

Comments to the authors:

The authors have addressed the comments, and the quality of the paper has met the requirements for publication. I recommend acceptance. This paper addresses the PU learning problem under structural label noise, and innovatively integrates the SAR assumption into a lightweight linear model to propose the P-MLR method. It conforms to

practical application scenarios and the experimental verification is relatively sufficient. After reading I still have the following questions and modification suggestions: 1.The proposed P-MLR method claims to achieve unbiased estimation of the positive class, is there any theoretical proof or justification for this? 2.The experimental section lacks key details for reproducibility, such as the training-validation set partitioning strategy and the number of model training epochs. There are some standardization issues in the paper. For example, SAR first appears in the abstract without its full English name.

ICIC 2026 Organization Committee

Sincerely Yours
ICIC
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